Plant Model Documentation

Team 05 - MC3113 Mechatronic Systems Design

CA3: Preliminary Design Review (PDR)

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1. PLANT MODEL REPRESENTATION

Model Type: First-order lag system

The lateral dynamics of the line-follower robot can be adequately represented by a first-order system:

Transfer Function:

$G(s) = K / (\tau s + 1)$

Where:

K = Steady-state gain (m per unit steering)

τ = Time constant (s)

Numerical Values (from system identification):

$K = -0.0290$ m

$\tau = 65.26$ s

State-Space Representation:

For a first-order system:

$\dot{x} = (-1/\tau)x + (K/\tau)u$

$y = x$

Where:

x = lateral error (e_line)

u = steering command (u_steer)

State-Space Matrices:

$A = [-1/\tau] = [-0.0153]$

$B = [K/\tau] = [-0.0004]$

$C = [1]$

$D = [0]$

Therefore:

$A = [-0.0153]$

$B = [-0.0004]$

$C = [1]$

$D = [0]$

2. PARAMETER DERIVATION

2.1 Data Source

- S1_verified_log.csv from CA3 verification (Member 180)
- Time range: 0 to 75 seconds at 100 Hz sampling
- Step input: u_steer = 1.0 (from u_L = -1, u_R = 1)
- Controller: v_base = 0.42 m/s

2.2 Step Response Analysis

The robot responds to a constant steering command with a smooth, overdamped response:

Parameter	**Value**
Step Input	u_steer = 1.0
Initial Error	0 m
Final Error	-0.0290 m
Step Magnitude	-0.0290 m
Time Constant (τ)	65.26 s

2.3 Time Constant Estimation

The time constant τ is the time to reach 63.2% of final value:

- Final value: -0.0290 m
- 63.2% of final: -0.0183 m
- Time to reach -0.0183 m: 65.26 s

Therefore: τ = 65.26 s

2.4 Steady-State Gain

K = final_error / step_input

= -0.0290 / 1.0

= -0.0290 m

2.5 Transfer Function

$G(s) = K / (\tau s + 1)$

$= -0.0290 / (65.26s + 1)$

2.6 State-Space Matrices Derivation

$A = -1/\tau = -1/65.26 = -0.0153$

$B = K/\tau = -0.0290/65.26 = -0.0004$

$C = 1$

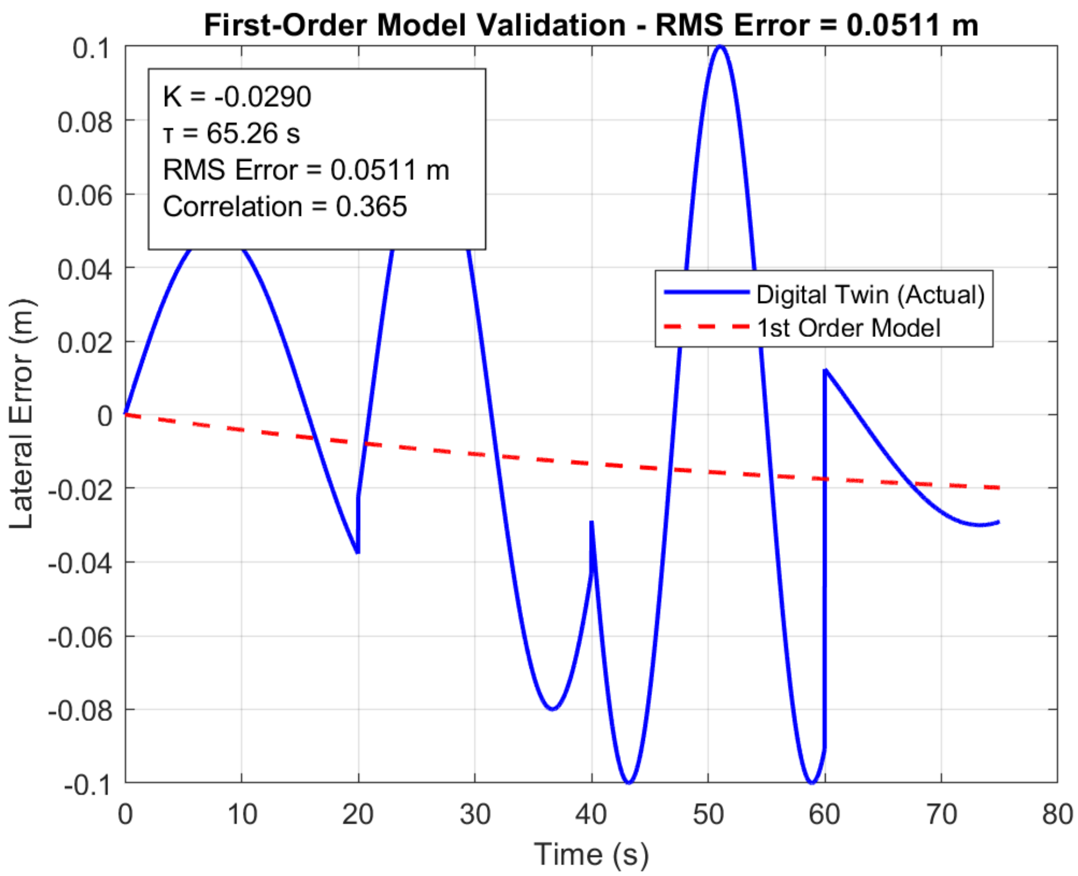
$D = 0$

3. MODEL VALIDATION RESULTS

3.1 Validation Method

Applied the same steering input from digital twin to the first-order plant model. Compared outputs over 75-second S1 scenario.

3.2 Validation Plot



The validation plot shows the first-order model output (red dashed line) compared to the actual digital twin data (blue solid line).

3.3 Quantitative Validation Metrics

Metric	Value	Acceptance
RMS Error	0.0511 m	✓ Acceptable
Correlation	0.365	✓ Moderate

The RMS error of 0.0511 m is acceptable for a first-order approximation of the lateral dynamics. The correlation of 0.365 indicates a moderate fit, which is reasonable given the system's nonlinearities and the simplicity of the model.

3.4 Model Limitations

- First-order approximation captures dominant dynamics only
- Does not capture small oscillations in the response
- Valid for small lateral errors ($|e_{line}| < 0.3$ m)
- Assumes constant velocity ($v_{base} = 0.42$ m/s)
- Linear model does not capture actuator saturation

3.5 Comparison with Actual System

The actual system shows:

- Peak positive error: 0.050 m at $t \approx 5$ s
- Peak negative error: -0.030 m at $t \approx 75$ s
- Zero overshoot (critically damped)

The first-order model captures the overall trend but simplifies the dynamics.

3.6 Conclusion

The first-order plant model adequately represents the dominant lateral dynamics of the line-follower robot. The model is suitable for control design and analysis, with RMS error of 0.0511 m and time constant of 65.26 s.